

The empirical recurrence for OEIS A202937, proved by transfer matrix

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Abstract

OEIS A202937 carries an empirical recurrence of order 10 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to a window of consecutive rows one longer than the bit-window the entry names – a vertical bit-window spans that many rows, so comparing two consecutive ones needs one row more – each of those rows being one of the few whose own bit-windows do not decrease along it, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 770$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A202937 is “Number of $(n+3) \times 9$ binary arrays with consecutive windows of four bits considered as a binary number nondecreasing in every row and column.”. It has offset 1 and begins

194481, 359970, 695114, 1360328, 2631416, 4958318, 9044252, 15949741, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = (1/15120)n^9 + (1/96)n^8 + (235/504)n^7 + (725/72)n^6 + 125n^5 + (276005/288)n^4 + (13991149/3024)n^3 + (1320845/72)n^2 + (24830539/420)n + 111295$. $a(n) = 10a(n-1) - 45a(n-2) + 120a(n-3) - 210a(n-4) + 252a(n-5) - 210a(n-6) + 120a(n-7) - 45a(n-8) + 10a(n-9) - a(n-10)$ for $n > 10$.

As of the “Last modified” line on the live entry (Jun 03 2018 07:57:43, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The condition the entry imposes is local: it constrains a window of consecutive rows one longer than the bit-window the entry names – a vertical bit-window spans that many rows, so comparing two consecutive ones needs one row more – each of those rows being one of the few whose own bit-windows do not decrease along it and nothing beyond. Take as vertices the admissible configurations of one such window and put an edge from u to v whenever v may follow u , which the condition decides by inspection of the two. An admissible array is then exactly a walk,

and ι, τ are the vectors recording which windows may begin and end an array, so with M the adjacency matrix on $S = 770$ vertices,

$$a(n) = \iota^\top M^n \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^{10} - \sum_i c_i t^{10-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+10} - \sum_i c_i M^{j+10-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 10a(n-1) - 45a(n-2) + 120a(n-3) - 210a(n-4) + 252a(n-5) - 210a(n-6) + 120a(n-7) - 45a(n-8) + 10a(n-9)$$

for every $n > 10$.

Proof. The vector $w = q(M)\tau$ was formed by 10 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 10 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.